

A Hierarchical Deep Video Understanding Method with Shot-Based Instance Search and Large Language Model

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ABSTRACT

Deep video understanding (DVU) is often considered a challenge due to the aim of interpreting a video with storyline, which is designed to solve two levels of problems: predicting the human interaction in scene-level and identifying the relationship between two entities in movie-level. Based on our understanding of the movie characteristics and analysis of DVU tasks, in this paper, we propose a four-stage method to solve the task, which includes video structuring, shot based instance search, interaction&relation prediction and shot-scene summary&Question Answering (QA) with ChatGPT. In these four stages, shot based instance search allows accurate identification and tracking of characters at an appropriate video granularity. Using ChatGPT in QA, on the one hand, can narrow the answer space, on the other hand, with the help of the powerful text understanding ability, ChatGPT can help us answer the questions by giving background knowledge. We rank first in movie-level group 2 and scene-level group 1, second in movie-level group 1 and scene-level group 2 in ACM MM 2023 Grand Challenge.

CCS CONCEPTS

• **Computing methodologies** → **Artificial intelligence**; *Knowledge representation and reasoning*;

KEYWORDS

video understanding, instance search, multi-modal feature

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1 INTRODUCTION

Deep video understanding is a difficult task which requires researchers to develop a deep analysis and understanding of the relationships between different entities in the video, to use known information to reason about other, and to populate a knowledge graph (KG) with all acquired information. The aim of the task is to push the limits of multimedia analysis techniques to address long duration videos holistically and extract useful knowledge to utilize it in solving different kinds of queries.

The DVU task tries to interpret a film from two levels: (1)scene-level: interactions between entities; (2)movie-level: relationships between entities. Instance identification and tracking is a very fundamental and important part of the solution to the different approaches to DVU. This is because many subsequent sessions are dependent on accurate instance search. For example, during the QA phase, a large number of questions focus on entities, like “What is the relationship between entity A to entity B?”. So you need to identify A and B precisely before you can proceed to the next process. Based on the above analysis, we believe that accurate identification of entities is the basis for problem solving. And in film, the core of the entity is person, because the narrative of the film is usually a human-centered story. So it is very crucial to do a good job of character identification and tracking.

According to our research on previous teams, we divided the teams into two groups according to the smallest unit for identification and tracking, one for long duration and one for short duration.

For the long duration group, [10, 12] perform character recognition and tracking under the scene. But doing so ignores an important feature of the movie, which is the shot. Unlike traditional face recognition and tracking under surveillance cameras, in which the scene is roughly the same, this aspect in movie encounters many problems. Because the film directors often use different camera language (express emotion or narrative through camera switching, advancement, and movement) to develop the story. But the switching camera makes it difficult to track the correct characters. For example, in the film often appear in the scene of two people talking face to face, the common camera switch is from a person’s face close-up switch to another, and in the switch before and after the two people in basically the same position, in this case, if you do

not consider the camera switch and directly track the frame of the series, it will be easy to make two different tracks in one trajectory.

For the short duration group, [15] samples every ten frames in the scene, and face detection and recognition in each frame, and by complementing the tracking track with the face track to obtain the complete entity track information. But this also does not take into account the misalignment of the trajectory brought about by the camera switch. And in shorter clips for identification and tracking, the character is likely to turn his back to the camera throughout the video, and we cannot discriminate the identity information of the tracking track through the face.

After considering the above two problems, we propose a character recognition tracking method for prestige of deep video understanding, which we call shot based instance search. In this method, we slice the scene into shots and perform face detection, recognition and entity tracking under the shots frame-by-frame. Then we determine the identity of the tracking by calculating the overlap between the face box and the tracking box, finally we vote in the same tracking track to get the identity of the whole track. In this way can we avoid the identity mismatch caused by camera switching, and also ensure through the voting mechanism that we can determine the identity information of the whole tracking sequence as long as the character has appeared facial information in one frame, even if the character turns around after.

In addition, due to the powerful understanding of text recently demonstrated by the large language model (LLM), we believe it is also very helpful for answering DVU questions, so we introduced ChatGPT into the QA phase of our model. In this phase, LLM assists us in two ways: on the one hand, we find that the class of entities has a great impact on the selection of relationship questions, so we use ChatGPT to classify all relationships, and on the other hand, to take full advantage of LLM's textual interpretation capabilities, we propose a method called shot-scene summary to obtain a scene description and let ChatGPT use this text as the basis for answering part of the scene-level questions and rank average it with the answers we obtained through the knowledge graph.

In general, our contribution is to:

- propose a character recognition and tracking method for movies, which can help us avoid tracking errors due to camera switching and achieve accurate instance search;
- introduce ChatGPT in QA phase. We use ChatGPT to help us classify different entity types and make multiple choice based on Shot-Scene Summary.

To test the effectiveness of our method, we use dataset and query from TRECVID 2022 DVU track [2] and ACM MM 2023 Grand Challenge and get good performance.

2 RELATED WORK

A two-stream network is proposed[14], which divides the network into Spatial Stream convolutional network and Temporal Stream convolutional network. After the two networks obtain classification scores about space and time respectively, the scores of the two streams are fused to obtain the final classification result. 3D convolutional network[16] is used to extract spatio-temporal information. On this basis, Pseudo-3D Residual Networks[11] are proposed to

decompose 3D convolutions, which reduced the overall parameter number of the model and made the calculation speed faster. It is proposed that the pre-trained 2D convolution kernel can be extended to 3D[3] with inflate, and dual-flow architecture can be applied in the network at the same time, and a frame of pictures can be copied in time. To get a boring video with exactly the same video frame, this approach allows the network to pre-train the model using a dataset of images. After that, the basic architecture of many researches is based on various variants of 3D-convolution. After transformer has made a great success in the field of NLP, people began to consider how to transfer transformer to the field of vision. [1] introduces the video classification architecture of pure transformer structure, and then encodes a series of transformer layers by mining spatio-temporal tokens. To solve the problem of video long distance dependence, a series of variant models are involved.

3 METHOD

In this section we introduce our four-stage method stage by stage, and our framework overview is shown in Fig. 1.

3.1 Video Structuring

The first step is video structuring. For each movie, we use Youtube ASR API to generate the subtitles. The Video Segmentation contains three parts, scene segmentation, shot segmentation and clip segmentation. For scene segmentation, we segment the movie using officially provided time stamps. For shot segmentation, we use PySceneDetect to segment the scene videos to facilitate the Shot Based Instance Search and Shot-Scene Summary. For clip segmentation, based on the subtitle file, the start and end time of each clip are obtained, and the corresponding scene to which clip belongs is located by combining the clip segmentation with the scene segmentation. For the clips which are between two segmented scenes, we add them to the scene which has a longer coincident video. Through this way can we generate the mapping table from scene to clip and the unit for the next process.

3.2 Shot Based Instance Search

The second step is the Shot Based Instance Search.

Person Recognition and Track For characters in shot, we run face detection using SCRFD [4] frame by frame to get bounding box and we use ArcFace [6] to extract the face features of snapshot and detected face. We average the different image features of the same person for identifying. Then we calculate the similarity between face library and detected face and save the similarity matrix to confirm the identity of the detected face. After face recognition, we run person track using faster RCNN [13] and Deepsort [17] on the shot-level to avoid tracking errors due to camera switching. Then we calculate the overlap between the detected face box and the tracking box. If the overlap exceeds the threshold, we assign face identity to the tracking frame. The identity information of the sequence is finally determined by voting. At last, we concatenate different sequences with same identity in all shots of one scene into one sequence.

Construction of Track Video Based on the character track sequence on the scene and the start and end time of the clip, the

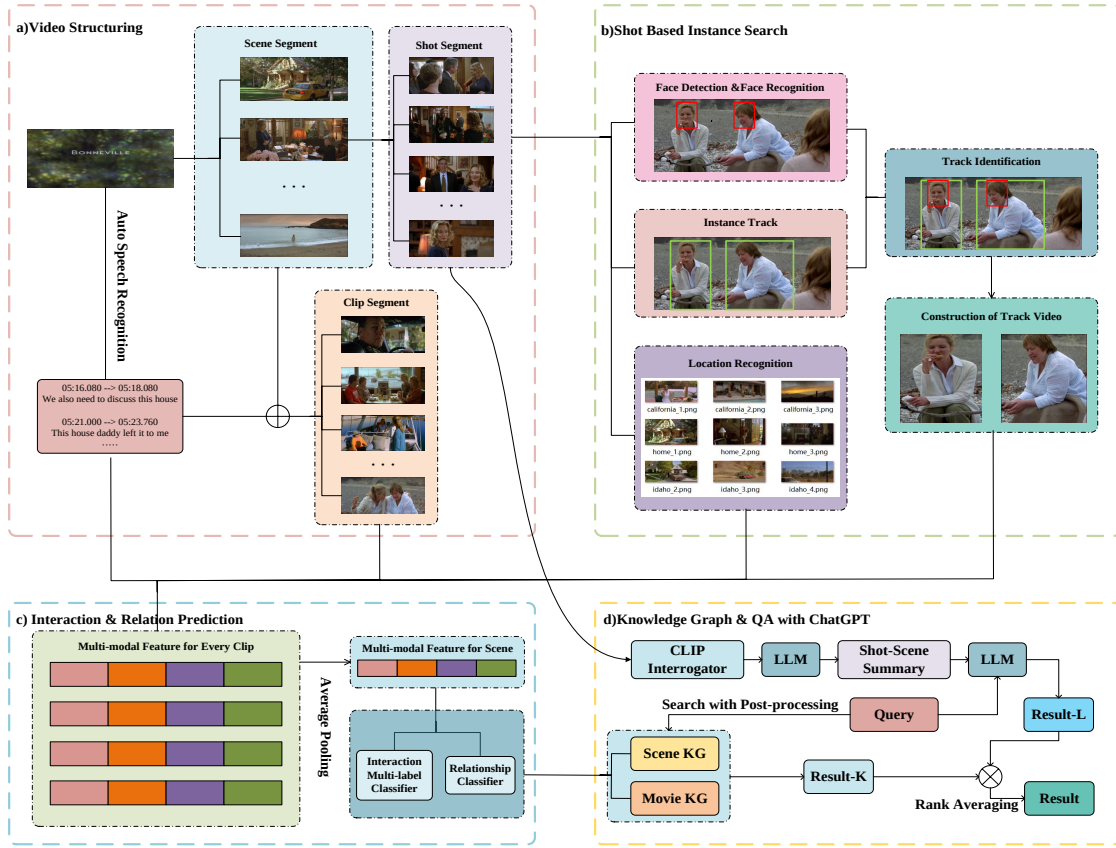


Figure 1: Our four-stage framework

track sequence on each clip is generated and saved as video for later usage.

Location Recognition The location recognition is similar to person recognition, we extract features by ResNet and average the features of different images for the same location. Then we extract the visual features of location by ResNet shot by shot, then we calculate the similarity with library to obtain the location identity in the shot.

3.3 Interaction Relation Prediction

The third step is the prediction of interaction and relation. We adopt a similar structure with Kukleva[8] for interaction recognition and relation recognition.

Multimodal Features We use multi-modal features for the prediction module, including visual features and text features.

Visual features We use TSM [9] to extract vectors of 2048 dimensions as visual features. For each clip, we extract features of every clip in a scene and for each combination of two track sequences. For example, if two persons and one location are recognized in the clip, there should be 6 different combinations of track features as there is active and passive relationships or interaction.

Text features For each clip, we use pre-trained BERT-base [7] to convert sentences to vectors of 768 dimensions as text features,

then we gather all the clip text feature belongs to the same scene. After extracting all the features in the scene, we concatenate the visual feature and text feature and get the feature of one scene.

Interaction and Relation Recognition Due to the annotations given officially are in scene-level rather than clip-level so we average the clip feature, which means every average scene feature may get multiple labels. Based on the multi-labeled feature, we apply the multi-label classification method to predict interaction between two persons. For the recognition of relation, we use the same method to get the average features as input to predict relationship.

3.4 Shot-Scene Summary & QA with ChatGPT

Knowledge Graph For each movie, we can get a list of predicted relations and interactions, then we use the Neo4j to generate the knowledge graph. The interactions and relations are saved in the different graphs. There are two kinds of nodes, the person node and location node. In interaction part, the connecting line is recognized interaction. In relation part, the connecting line is relation.

Shot-Scene Summary In order to use ChatGPT, we need to provide it with background knowledge in textual form to help us answer the questions, so we propose a shot-scene summary approach. We use CLIP Interrogator to generate picture descriptions for the middle frame of each shot in a scene, because the middle part

is often the richest part of a movie information, and after obtaining descriptions of all the shots, these descriptions are stitched together and fed to ChatGPT, allowing it to remove duplicate semantics and regenerate a more refined description for after usage.

QA with ChatGPT In ACM MM 2023 Grand Challenge, there are 5 types of question, in scene-level:

- **Find the unique scene (s1).** Base on the sequence of actions to find the correct scene.
- **Fill in the graph space (s2).** Find the person in a specific scene with list of person and interactions.
- **Find next or previous interaction (s3).** Find next/previous interaction between person based on interactions sequence in a scene.
- **Find the 1-to-1 relationship between scenes and natural language descriptions (s4).** Match the correct natural language description for each scene.
- **Classify scene sentiment from a given scene (s5).** Classify the correct sentiment label for each scene.

In movie-level, there are also 2 types of question:

- **Fill in the graph space (s1).** Find the entity based on the relation with others.
- **Question Answering (s2).** Find the choice based on the question.

For s1 and s2 in scene-level, we transfer the query to the Cypher language, and search in the KG. Finally we get a set of scenes, and we take the top four as our answers. For s3, we have two ways to solve it. One is similar to s1, first to the Cypher language and then search in KG to get answer. And the other is through LLM, we input the query and shot-scene summary as background, have ChatGPT rank all options. Then we rank average the answers we gave with the answers given by ChatGPT to get the final answer. In the experiment, we assigned the score of the six answers as 6,5,4,3,2,1 by order, and we think that the accuracy of shot-scene summary and the interpretation ability of ChatGPT are very strong, so we assign 1 to the weight of the answers generated by ChatGPT, the answers we obtained through the KG were assigned a weight of 0.6. For s1 in movie-level, we solve them by finding the relation line between the character nodes mentioned in the queries. For s2, we solve them by finding the relation line that satisfy the right entity nodes in the graph, specially, we find that there are different relationship libraries between different categories of entities. We use ChatGPT to divide the relation terms into two classes *e.g.*, PP (person-to-person) and PL (person-to-location) to reduce the answer space.

4 EXPERIMENTS

4.1 Dataset & Query

We use the dataset from NIST [5], which include 19 training movies (total 25 hr) and corresponding annotations, snapshots, vocabulary and, 5 testing movies (total 7.5 hr) with snapshots and corresponding query from ACM MM 2023 Grand Challenge, and use **MRR** and **ACC** as metrics. In addition, we also used the relevant data and query of NIST TRECVID 2022 DVU task as test in early stage.

4.2 Evaluation and Results

The results in NIST TRECVID 2022 DVU task and ACM MM 2023 Grand Challenge are listed in Table 1 (only a part of question types we mentioned above) and 2.

Table 1: result in NIST TRECVID 2022 DVU task(%)

Movie	Scene-level		Movie-level	
	s1(MRR)	s3(ACC)	s1(MRR)	s2(ACC)
Calloused_Hands	6.25	37.5	62.5	16.6
Chained_For_Life	25	0	66.7	66.7
Liberty_Kid	8.32	37.5	50	50
Like_Me	25	37.5	100	66.7
Little_Rock	0	0	37.5	28.5
Losing_Ground	8.32	50	60	50
total	22.0		47.1	

Table 2: result in ACM MM 2023 Grand Challenge(%)

Movie	Scene-level					Movie-level	
	s1(MRR)	s2(MRR)	s3(ACC)	s4(ACC)	s5(ACC)	s1(MRR)	s2(ACC)
Archipelago	12.5	87.5	62.5	40	33.3	81.25	33.3
Bonneville	37.5	100	25	50	66.7	62.5	39.1
Heart_Machine	25	100	50	60	33.3	37.5	61.5
Little_Rock	18.8	57.5	0	60	33.3	62.5	30
Memphis	8.32	100	12.5	60	66.7	54.2	59.1
total	40.9		51.2			59.6	43.0

Through the analysis of different question and answers, we found that our model has a better performance in the questions that Fill in the graph space. As we mentioned above, this type of questions has strong requirements for the recognition of characters, and our method is accurate in the recognition of characters. But we also found some queries that we didn't do well. The first problem is the data distribution of the training set is not even (long-tail), our method will tend to choose the answer with more training samples. The second problem is indistinguishable action by visual aspect alone, such as "talk to" and "asks". Our method has difficulty distinguishing similar pairs of these answers. We also tried to add audio features to assist this problem, but it did not work. We believe that the dimension of some commonly used audio features (MFCC, etc.) is small (less than 100). Compared with the text-visual features (6912 dimensions), audio features can provide little information.

5 CONCLUSION

In this paper, we propose a hierarchical DVU method with shot-based instance search and large language model. We introduce shot-based instance search to cope with track error, and we also use LLM to answer questions for us by providing background knowledge for ChatGPT. We have evaluated our method in the ACM MM 2023 Grand Challenge, and obtain competitive results.

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